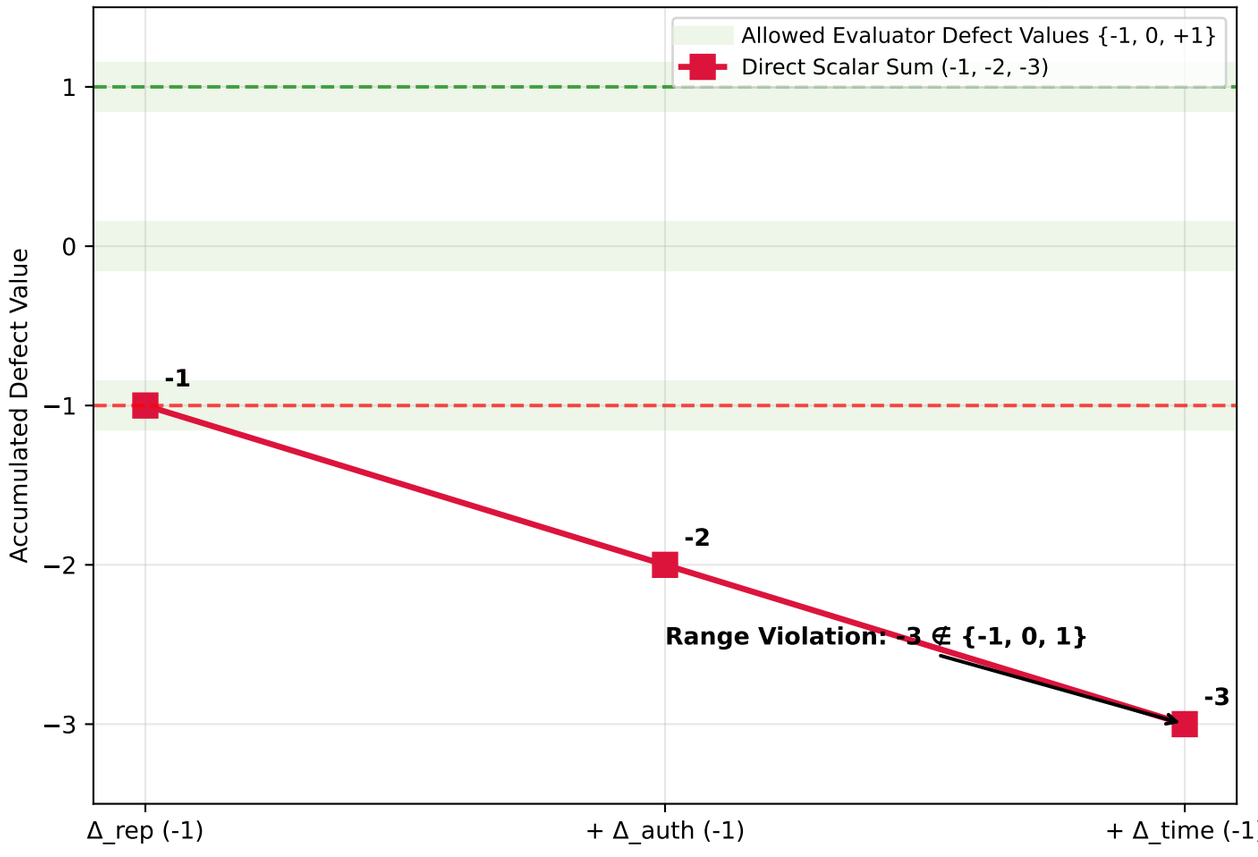
Local Green, Global Red II: Organizational Evaluator Gluing

A. Evaluator Gluing Defect Trichotomy Δ_E

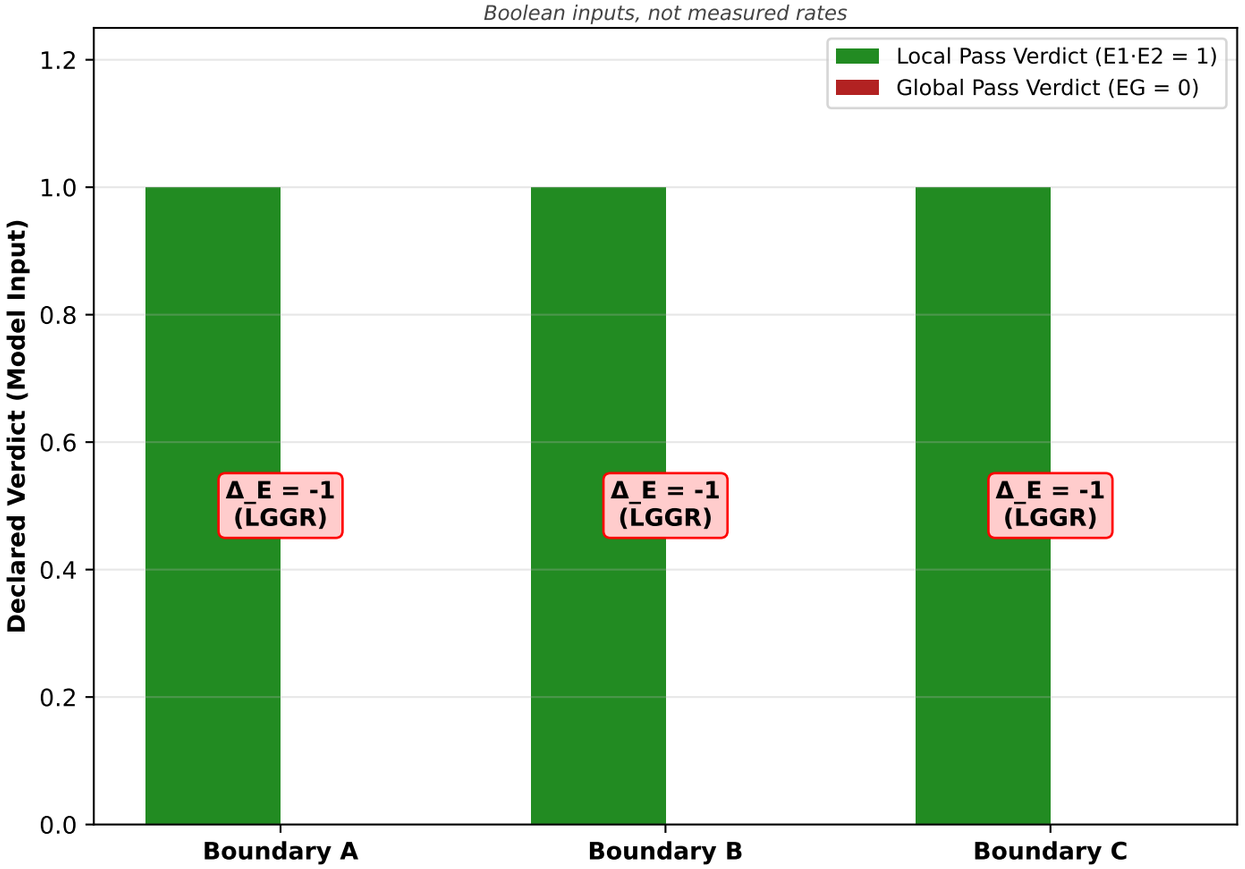
E1	E2	∏ Ei	EG	Δ_E	Regime
0	0	0	0	0	Agreement at Joint Failure
0	1	0	0	0	Agreement (Partial Fail)
1	0	0	0	0	Agreement (Partial Fail)
1	1	1	0	-1	Disagreement: Local Green, Global Red (LGGR)
0	0	0	1	+1	Disagreement: Global-Only Pass
0	1	0	1	+1	Disagreement: Global-Only Pass
1	0	0	1	+1	Disagreement: Global-Only Pass
1	1	1	1	0	Agreement at Joint Pass

B. Direct sums can leave the Boolean defect range



C. Three local-green / global-red assignments

Boolean inputs, not measured rates



D. Boundary Governance & Verification Architecture

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Boolean boundary model:

  Evaluator Defect: Δ_E = E_G - ∏ E_i ∈ {-1, 0, +1}
  Factorization:    E_G = ∏ E_i  (Δ_E = 0)

Declared governance checks:

  • G_rep  (Representational sufficiency)
  • G_auth (authority + accountability)
  • G_time (Temporal coherence)

Weak-sufficiency counterexample (Proposition 2):

  G_rep ∧ G_auth ∧ G_time DOES NOT FORCE E_G = ∏ E_i
  (Counterexample: R recovers E_G, yet Δ_E = -1)

Checked-closure eligibility (definition):

  Requires BOTH evaluator agreement (Δ_E = 0)
  AND verified governance prerequisites.
  GovernanceReady ==> G_auth  [conjunction elimination]

Detector note:

  • RED (Δ_E = -1): Sound detection of non-factorization
  • GREEN (Δ_E = 0): Unsound certificate without adequacy
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